# Complete List of Peer-Reviewed Publications

## Publication Order


[1] A. Dujakovich, B. M. Walsh, G. Cucho-Padin, E. Atz, K. Gendreau, J. Hare, C. B. Markwardt, C. O'Brien, F. S. Porter, **R. Qudsi**, D. G. Sibeck, "Exploring a Terrestrial X-Ray Source and Auroral Emissions Using NICER," *Journal of Geophysical Research (Space Physics)*, vol. 131, no. 2, e2025JA034213, e2025JA034213, Feb. 2026. DOI: 10.1029/2025JA034213.

[2] Carson P. Brown, Bennett A. Maruca, Vaishali Prabakaran, **Ramiz A. Qudsi**, B. M. Walsh, B. L. Alterman, Philip A. Isenberg, "Evaluating the Parker Model With the Trans-Heliospheric Survey," *Geophysical Research Letters*, vol. 52, no. 12, e2025GL115186, e2025GL115186, Jun. 2025. DOI: 10.1029/2025GL115186.

[3] Catriana K. Paw U, Brian Walsh, **Ramiz Qudsi**, Meghan LeMay, Cadin Connor, Luisa Capannolo, Sam Busk, Rousseau Nutter, Kip D. Kuntz, Frederick Scott Porter, Hyunju K. Connor, Dennis Chornay, "Magnetic charged particle diverters' performance across space plasma environments," *Journal of Astronomical Telescopes, Instruments, and Systems*, vol. 11, 044006, p. 044 006, Oct. 2025. DOI: 10.1117/1.JATIS.11.4.044006.

[4] Hyangpyo Kim, Hyunju K. Connor, Jaewoong Jung, Brian M. Walsh, David Sibeck, Kip D. Kuntz, Frederick S. Porter, Catriana K. Paw U, Rousseau A. Nutter, **Ramiz Qudsi**, Rumi Nakamura, Michael Collier, "Estimating the subsolar magnetopause position from soft X-ray images using a low-pass image filter," *Earth and Planetary Physics*, vol. 8, no. 1, pp. 173–183, Jan. 2024. DOI: 10.26464/epp2023069.

[5] C. O'Brien, B. M. Walsh, Y. Zou, **R. Qudsi**, S. Tasnim, H. Zhang, D. G. Sibeck, "PRIME-SH: A Data-Driven Probabilistic Model of Earth's Magnetosheath," *Journal of Geophysical Research Machine Learning and Computation*, vol. 1, no. 3, e2024JH000235, e2024JH000235, Sep. 2024. DOI: 10.1029/2024JH000235.

[6] Catriana K. Paw U, Brian M. Walsh, **Ramiz Qudsi**, Sam Busk, Cadin Connor, Dennis Chornay, Hyunju K. Connor, Kip D. Kuntz, Rousseau Nutter, F. Scott Porter, "Simulation of the charged particle deflection from the sweeping magnet array in the Lunar Environment heliospheric X-ray imager," *Review of Scientific Instruments*, vol. 95, no. 12, 123301, p. 123 301, Dec. 2024. DOI: 10.1063/5.0230759.

[7] B. M. Walsh, K. D. Kuntz, S. Busk, T. Cameron, D. Chornay, A. Chuchra, M. R. Collier, C. Connor, H. K. Connor, T. E. Cravens, N. Dobson, M. Galeazzi, H. Kim, J. Kujawski, C. K. Paw U, F. S. Porter, V. Naldoza, R. Nutter, **R. Qudsi**, D. G. Sibeck, S. Sembay, M. Shoemaker, K. Simms, N. E. Thomas, E. Atz, G. Winkert, "The Lunar Environment Heliophysics X-ray Imager (LEXI) Mission," *Space Science Reviews*, vol. 220, no. 4, 37, p. 37, May 2024. DOI: 10.1007/s11214-024-01063-4.

[8] E. Johnson, B. A. Maruca, M. McManus, K. G. Klein, E. R. Lichko, J. Verniero, K. W. Paulson, H. DeWeese, I. Dieguez, **R. A. Qudsi**, J. Kasper, M. Stevens, B. L. Alterman, L. B. Wilson, R. Livi, A. Rahmati, D. Larson, "Anterograde Collisional Analysis of Solar Wind Ions," *The Astrophysical Journal*, vol. 950, no. 1, 51, p. 51, Jun. 2023. DOI: 10.3847/1538-4357/accc32.

[9] Bennett A. Maruca, **Ramiz A. Qudsi**, B. L. Alterman, Brian M. Walsh, Kelly E. Korreck, Daniel Verscharen, Riddhi Bandyopadhyay, Rohit Chhiber, Alexandros Chasapis, Tulasi N. Parashar, William H. Matthaeus, Melvyn L. Goldstein, "The Trans-Heliospheric Survey. Radial trends in plasma parameters across the heliosphere," *Astronomy and Astrophysics*, vol. 675, A196, A196, Jul. 2023. DOI: 10.1051/0004-6361/202345951.

[10] **Ramiz A. Qudsi**, Brian M. Walsh, Jeff Broll, Emil Atz, Stein Haaland, "Statistical Comparison of Various Dayside Magnetopause Reconnection X-Line Prediction Models," *Journal of Geophysical Research (Space Physics)*, vol. 128, no. 10, e2023JA031644, e2023JA031644, Oct. 2023. DOI: 10.1029/2023JA031644.

[11] Riddhi Bandyopadhyay, **Ramiz A. Qudsi**, S. Peter Gary, William H. Matthaeus, Tulasi N. Parashar, Bennett A. Maruca, Vadim Roytershteyn, Alexandros Chasapis, Barbara L. Giles, Daniel J. Gershman, Craig J. Pollock, Christopher T. Russell, Robert J. Strangeway, Roy B. Torbert, Thomas E. Moore, James L. Burch, "Interplay of turbulence and proton-microinstability growth in space plasmas," *Physics of Plasmas*, vol. 29, no. 10, 102107, p. 102 107, Oct. 2022. DOI: 10.1063/5.0098625. arXiv: 2006.10316 [physics.space-ph].

[12] Haley DeWeese, Bennett A. Maruca, **Ramiz A. Qudsi**, Alexandros Chasapis, Mark Pultrone, Elliot Johnson, Sarah K. Vines, Michael A. Shay, William H. Matthaeus, Roman G. Gomez, Stephen A. Fuselier, Barbara L. Giles, Daniel J. Gershman, Christopher T. Russell, Robert J. Strangeway, James L. Burch, Roy B. Torbert, "Alpha Particle Temperature Anisotropy in Earth's Magnetosheath," *The Astrophysical Journal*, vol. 941, no. 1, 12, p. 12, Dec. 2022. DOI: 10.3847/1538-4357/ac9791.

[13] Nikos Sioulas, Zesen Huang, Marco Velli, Rohit Chhiber, Manuel E. Cuesta, Chen Shi, William H. Matthaeus, Riddhi Bandyopadhyay, Loukas Vlahos, Trevor A. Bowen, **Ramiz A. Qudsi**, Stuart D. Bale, Christopher J. Owen, P. Louarn, A. Fedorov, Milan Maksimović, Michael L. Stevens, Anthony Case, Justin Kasper, Davin Larson, Marc Pulupa, Roberto Livi, "Magnetic Field Intermittency in the Solar Wind: Parker Solar Probe and SolO Observations Ranging from the Alfvén Region up to 1 AU," *The Astrophysical Journal*, vol. 934, no. 2, 143, p. 143, Aug. 2022. DOI: 10.3847/1538-4357/ac7aa2. arXiv: 2206.00871 [astro-ph.SR].

[14] Nikos Sioulas, Marco Velli, Rohit Chhiber, Loukas Vlahos, William H. Matthaeus, Riddhi Bandyopadhyay, Manuel E. Cuesta, Chen Shi, Trevor A. Bowen, **Ramiz A. Qudsi**, Michael L. Stevens, Stuart D. Bale, "Statistical Analysis of Intermittency and its Association with Proton Heating in the Near-Sun Environment," *The Astrophysical Journal*, vol. 927, no. 2, 140, p. 140, Mar. 2022. DOI: 10.3847/1538-4357/ac4fc1. arXiv: 2201.10067 [astro-ph.SR].

[15] C. L. Lentz, A. Chasapis, **R. A. Qudsi**, J. Halekas, B. A. Maruca, L. Andersson, D. N. Baker, "On the Solar Wind Proton Temperature Anisotropy at Mars' Orbital Location," *Journal of Geophysical Research (Space Physics)*, vol. 126, no. 10, e29438, e29438, Oct. 2021. DOI: 10.1029/2021JA02943810.1002/essoar.10506890.1.

[16] Bennett A. Maruca, Jeffersson A. Agudelo Rueda, Riddhi Bandyopadhyay, Federica B. Bianco, Alexandros Chasapis, Rohit Chhiber, Haley DeWeese, William H. Matthaeus, David M. Miles, **Ramiz A. Qudsi**, Michael J. Richardson, Sergio Servidio, Michael A. Shay, David Sundkvist, Daniel Verscharen, Sarah K. Vines, Joseph H. Westlake, Robert T. Wicks, "MagneToRE: Mapping the 3-D Magnetic Structure of the Solar Wind Using a Large Constellation of Nanosatellites," *Frontiers in Astronomy and Space Sciences*, vol. 8, 108, p. 108, Jul. 2021. DOI: 10.3389/fspas.2021.665885.

[17] **Ramiz Ahmad Qudsi**, "On the Interplay between Microkinetics and Turbulence in Space Plasmas," Ph.D. dissertation, University of Delaware, Department of Physics and Astronomy, Jan. 2021.

[18] Riddhi Bandyopadhyay, M. L. Goldstein, B. A. Maruca, W. H. Matthaeus, T. N. Parashar, D. Ruffolo, R. Chhiber, A. Usmanov, A. Chasapis, **R. Qudsi**, Stuart D. Bale, J. W. Bonnell, Thierry Dudok de Wit, Keith Goetz, Peter R. Harvey, Robert J. MacDowall, David M. Malaspina, Marc Pulupa, J. C. Kasper, K. E. Korreck, A. W. Case, M. Stevens, P. Whittlesey, D. Larson, R. Livi, K. G. Klein, M. Velli, N. Raouafi, "Enhanced Energy Transfer Rate in Solar Wind Turbulence Observed near the Sun from Parker Solar Probe," *The Astrophysical Journal Supplement Series*, vol. 246, no. 2, 48, p. 48, Feb. 2020. DOI: 10.3847/1538-4365/ab5dae. arXiv: 1912.02959 [physics.space-ph].

[19] Riddhi Bandyopadhyay, W. H. Matthaeus, T. N. Parashar, R. Chhiber, D. Ruffolo, M. L. Goldstein, B. A. Maruca, A. Chasapis, **R. Qudsi**, D. J. McComas, E. R. Christian, J. R. Szalay, C. J. Joyce, J. Giacalone, N. A. Schwadron, D. G. Mitchell, M. E. Hill, M. E. Wiedenbeck, R. L. McNutt, M. I. Desai, Stuart D. Bale, J. W. Bonnell, Thierry Dudok de Wit, Keith Goetz, Peter R. Harvey, Robert J. MacDowall, David M. Malaspina, Marc Pulupa, M. Velli, J. C. Kasper, K. E. Korreck, M. Stevens, A. W. Case, N. Raouafi, "Observations of Energetic-particle Population Enhancements along Intermittent Structures near the Sun from the Parker Solar Probe," *The Astrophysical Journal Supplement Series*, vol. 246, no. 2, 61, p. 61, Feb. 2020. DOI: 10.3847/1538-4365/ab6220. arXiv: 1912.03424 [physics.space-ph].

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[21] S. Peter Gary, Riddhi Bandyopadhyay, **Ramiz A. Qudsi**, William H. Matthaeus, Bennett A. Maruca, Tulasi N. Parashar, Vadim Roytershteyn, "Particle-in-cell Simulations of Decaying Plasma Turbulence: Linear Instabilities versus Nonlinear Processes in 3D and 2.5D Approximations," *The Astrophysical Journal*, vol. 901, no. 2, 160, p. 160, Oct. 2020. DOI: 10.3847/1538-4357/abb2ac.

[22] Jia Huang, J. C. Kasper, D. Vech, K. G. Klein, M. Stevens, Mihailo M. Martinović, B. L. Alterman, Tereza Ďurovcová, Kristoff Paulson, Bennett A. Maruca, **Ramiz A. Qudsi**, A. W. Case, K. E. Korreck, Lan K. Jian, Marco Velli, B. Lavraud, A. Hegedus, C. M. Bert, J. Holmes, Stuart D. Bale, Davin E. Larson, Roberto Livi, P. Whittlesey, Marc Pulupa, Robert J. MacDowall, David M. Malaspina, John W. Bonnell, Peter Harvey, Keith Goetz, Thierry Dudok de Wit, "Proton Temperature Anisotropy Variations in Inner Heliosphere Estimated with the First Parker Solar Probe Observations," *The Astrophysical Journal Supplement Series*, vol. 246, no. 2, 70, p. 70, Feb. 2020. DOI: 10.3847/1538-4365/ab74e0. arXiv: 1912.03871 [physics.space-ph].

[23] T. N. Parashar, M. L. Goldstein, B. A. Maruca, W. H. Matthaeus, D. Ruffolo, R. Bandyopadhyay, R. Chhiber, A. Chasapis, **R. Qudsi**, D. Vech, D. A. Roberts, S. D. Bale, J. W. Bonnell, T. Dudok de Wit, K. Goetz, P. R. Harvey, R. J. MacDowall, D. Malaspina, M. Pulupa, J. C. Kasper, K. E. Korreck, A. W. Case, M. Stevens, P. Whittlesey, D. Larson, R. Livi, M. Velli, N. Raouafi, "Measures of Scale-dependent Alfvénicity in the First PSP Solar Encounter," *The Astrophysical Journal Supplement Series*, vol. 246, no. 2, 58, p. 58, Feb. 2020. DOI: 10.3847/1538-4365/ab64e6.

[24] **R. A. Qudsi**, B. A. Maruca, W. H. Matthaeus, T. N. Parashar, Riddhi Bandyopadhyay, R. Chhiber, A. Chasapis, Melvyn L. Goldstein, S. D. Bale, J. W. Bonnell, T. Dudok de Wit, K. Goetz, P. R. Harvey, R. J. MacDowall, D. Malaspina, M. Pulupa, J. C. Kasper, K. E. Korreck, A. W. Case, M. Stevens, P. Whittlesey, D. Larson, R. Livi, M. Velli, N. Raouafi, "Observations of Heating along Intermittent Structures in the Inner Heliosphere from PSP Data," *The Astrophysical Journal Supplement Series*, vol. 246, no. 2, 46, p. 46, Feb. 2020. DOI: `10.3847/1538-4365/ab5c19`. arXiv: `1912.05483 [physics.space-ph]`.

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Complete list of publication: **ADS**, **Google Scholar**